

CHAPTER ONE: INTRODUCTION

BASIC TERMINOLOGIES

System : It is a combination or an arrangement of different physical components which act together as an entire unit to achieve a primary objective

Examples of Systems: Human beings, mechanical devices, an electrical switch, plants, animals, the atmosphere, the stock market, the political systems, e.t.c.

A system Can be physical, ecological, biological, etc.

Control: The action to command, direct or regulate a system

Control System: It is an arrangement of different physical elements in such a manner so as to regulate, direct or command itself or some other system. For example, if a lamp is switched ON or OFF using a switch, the entire system can be called a control system.

Plant or Process: The part or component of a system that is required to be controlled

Controller/Control element: The element of the system itself or external to the system which controls the plant or the process

Input/Controlling Variable/command input: It is an applied signal or an excitation signal applied to a control system from an external energy source in order to produce a specified output

Output/Controlled Variable: It is the actual response obtained from a control system when input is applied to it.

Disturbance: It is a signal which tends to adversely affect the value of the output of a system. If such a disturbance is generated within a system, it is called an **internal disturbance**. The disturbance generated outside the system, acting as an extra input to the system in addition to its normal input, affecting the output adversely is called an **external disturbance**.

Actuator: It is the device that causes the process to provide the output. It is the device that provides the motive power to the process.

Multivariable control system: A system with more than one input variable or more than one output variable.

Reference Input: It is the actual signal input to a control system.

Actuating Signal/Error Signal: It is the difference between the reference input and feedback signal.

CLASSIFICATION OF CONTROL SYSTEMS

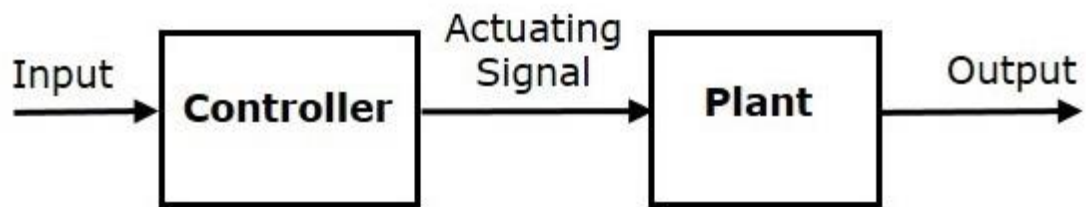
Control systems are broadly classified as:

1. **Natural Control System:** It is a control system that is created by nature i.e. solar system, digestive system of any animal, respiration system, etc.

2. **Man-made control system:** It is a control system that is created by humans i.e. automobiles, power plants, switches, etc.
3. **Combinational/Hybrid control system:** It is a control system that is a combination of natural and man-made control systems e.g. driver driving a vehicle.
4. **Time varying and Time-Invariant systems:** Time-variant control system is a system where any one or more parameters of the control system vary with time i.e. driving a vehicle. Time-invariant control system is where none of the parameters of the control system vary with time i.e. control system made up of inductors, capacitors and resistors only
5. **Open-Loop control system and closed-loop control system:** Open-loop control system is a control system where its control action only depends on its input signal and does not depend on its output response. Closed-loop control system is a control system where its control action depends on both of its input signal and output response.

OPEN LOOP CONTROL SYSTEM

The following figure shows the block diagram of the open loop control system



Reference input, $r(t)$ is applied to the controller which generates the actuating signal (u) required to control the process which is to be controlled. Process is giving out the necessary desired controlled output $c(t)$

Advantages of Open-loop Control System

- i. Simple to construct
- ii. Simple to design and hence economical
- iii. Highly stable operation
- iv. Convenient when output is difficult to measure

Disadvantages of Open-loop Control System

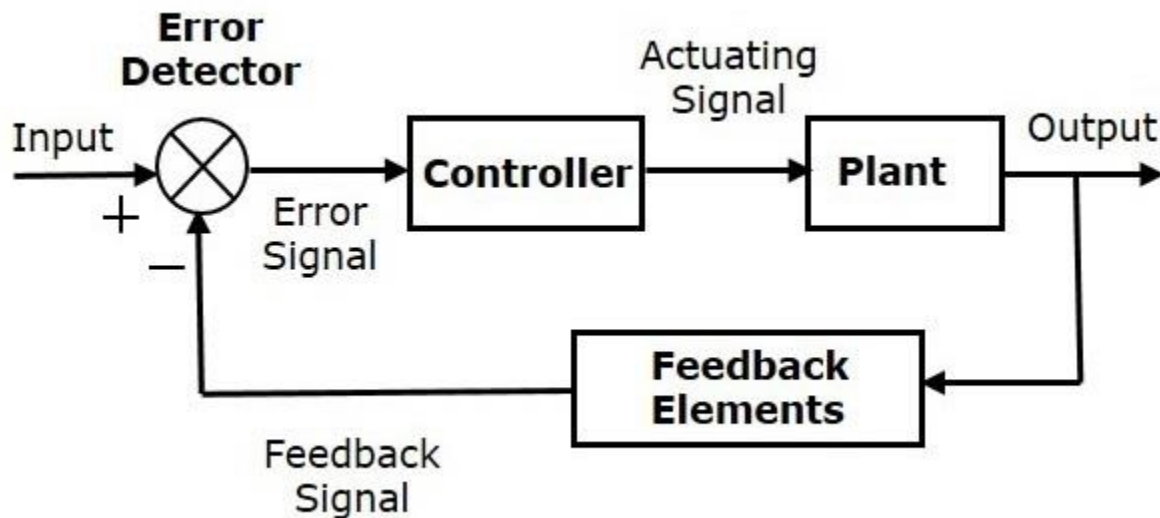
- i. Inaccurate and unreliable when input or system parameters are variable in nature
- ii. Recalibration of the parameters are required time to time to maintain quality and accuracy

- iii. They cannot sense internal disturbances in the system after controller stage

Examples of open-loop control systems are: traffic signal, washing machine, bread toaster, electric switch, etc.

CLOSED-LOOP CONTROL SYSTEM

The following figure shows the block diagram of a closed loop control system.



The error detector produces an error signal, which is the difference between the input and the feedback signal. This feedback signal is obtained from the block (feedback elements) by considering the output of the overall system as an input to this block. Instead of the direct input, the error signal is applied as an input to a controller.

So, the controller produces an actuating signal which controls the plant. In this combination, the output of the control system is adjusted automatically till we get the desired response. Hence, the closed loop control systems are also called the automatic control systems

It is a control system where its control action depends on both its input signal and output response.

Examples: automatic electric iron, missile launcher, speed control of dc motor, etc

Advantages of Closed-loop control system

- i. High bandwidth of operation
- ii. Less nonlinearity effect of these systems on output response
- iii. Highly accurate than open-loop systems
- iv. Can operate efficiently when input or system parameters are variable in nature
- v. There is facility of automation

Disadvantages of Closed-loop control system

- i.** Complex design and difficult to construct
- ii.** Expensive than that of open-loop control system
- iii.** Complicated for maintenance
- iv.** Less stable operation than that of open-loop control system

Comparison between open-loop and closed-loop control systems

S/NO.	OPEN-LOOP CONTROL SYSTEM	CLOSED-LOOP CONTROL SYSTEM
1	No feedback is given to the control system	A feedback is given to the control system
2	There is a possibility of undesirable system oscillations(hunting)	Introduces the possibility of undesirable system oscillation(hunting)
3	Cannot be intelligent	Intelligent and controlling action
4	The output will not vary for a constant input, provided the system parameters remain unaltered	In the system, the output may vary for a constant input, depending upon the feedback
5	Error detection is absent	Error detection is present
6	Small bandwidth	Large bandwidth
7	More stable	Less stable or prone to instability
8	Affected by non-linearities	Not affected by non-linearities
9	Very sensitive in nature	Less sensitive to disturbances
10	Simple design	Complex design
11	Cheap	Costly